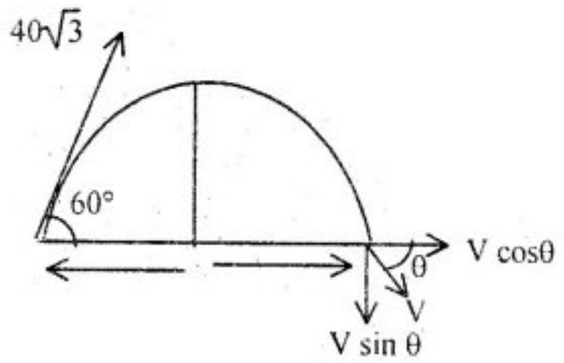
Exercise 19.4

1. A particle projected upwards from the level ground at an angle of 60° with the horizon has an initial speed of $40\sqrt{3}$ ms^{-1}. ($g = 10\ ms^{-1}$)

(a) How long will it be before it hits the ground ?

(b) How far from the starting point will it strike ?



Sol^n: initial velocity (u) = $40\sqrt{3}$ ms^{-1}

angle of projection (α) = 60°

(a) Let, T be the time of flight

$$\therefore \quad T = \frac{2u\sin\alpha}{g} = \frac{2 \times 40\sqrt{3} \times \sin 60°}{10} = 2 \times 4\sqrt{3} \times \frac{\sqrt{3}}{2} = 12 \text{ secs.}$$

(b) Let, H he the horizontal range

$$\therefore \quad H = \frac{u^2 \sin 2\alpha}{g} = \frac{(40\sqrt{3})^2 \times \sin(2 \times 60°)}{10} = \frac{40\sqrt{3} \times 40\sqrt{3}}{10} \times \frac{\sqrt{3}}{2} = 240\sqrt{3} \text{ m.}$$

2. A ball is projected with an initial upward velocity component of 20 ms^{-1} and a horizontal velocity component of 25 ms^{-1}. ($g = 9.8\ ms^{-2}$)

(a) Find the position and the velocity of the ball after (i) 2 s (ii) 3 s (iii) 4 s.

(b) How much time is required to reach the highest point of the trajectory ?

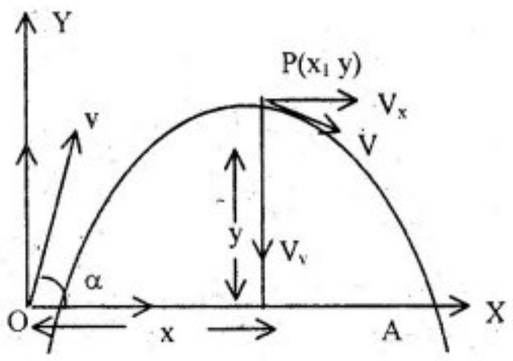
(c) How high is the point?

(d) How much time (after launch) is required for the ball to return to its original level?

(e) How far has it travelled horizontally during this time?

Illustrate your answer with neat and sufficiently large sketch.

Sol^n: (a) Let, V be the velocity and P(x, y) be position of projectile at time t. Let, v be the velocity and α be angle of projection



$v \cos\alpha = 25$, $v \sin\alpha = 20$

$x = (v \cos\alpha)$ ..... (i)

$y = (v \sin\alpha)t - \frac{1}{2}gt^2$ ..... (ii)

Then V is the resultant of two velocities V_x and V_y given by

$V_x = v \cos\alpha$ (along horizontal) ..... (iii)

and $V_y = v \sin\alpha - gt$ (along vertical) ..... (iv)

(i) After 2 second

Substituting value of t = 2, on (i), (ii), (iii) and (iv), we get

$x = 25 \times 2 = 50$ m.

$y = 20 \times 2 - \frac{1}{2} \times 9.8 \times 2^2 = 40 - 19.6 = 20.4$ m

$V_x = 25\ ms^{-1}$ Ans

$V_y = 20 - 9.8 \times 2 = 20 - 19.6 = 0.40\ ms^{-1}$

(ii) After 3 seconds

Substituting the value of t = 3 on (i), (ii), (iii) and (iv), we get

$x = 25 \times 3 = 75$ m

$y = 20 \times 3 - \frac{1}{2} \times 9.8 \times 3^2 = 60 - 44.1 = 15.9$ m

$V_x = 25\ ms^{-1}$

$V_y = 20 - 9.8 \times 3 - 9.4\ ms^{-1}$

(iii) After 4 seconds

Substituting the value of t = 4 on (i), (ii), (iii) and (iv), we get

$x = 25 \times 4 = 100$ m

$y = 20 \times 4 - \frac{1}{2} \times 9.8 \times 4^2 = 80 - 78.4 = 1.6$ m

$V_x = 20\ ms^{-1}$

$V_y = 20 - 9.8 \times 4 = 20 - 39.2 = -19.2\ ms^{-1}$

(b) Let, t be required time to reach the highest point, then $t = \frac{v \sin\alpha}{g} = \frac{20}{9.8} = 2.04$ sec.

(c) Let, H is the greatest height attained by the ball, then

$H = \frac{v^2 \sin^2\alpha}{2g} = \frac{(v \sin\alpha)^2}{2g} = \frac{20^2}{2 \times 9.8} = \frac{400}{19.6} = 20.4$ m.

(d) Let T be the time of flight, then

$T = \frac{2v \sin\alpha}{g} = \frac{2 \times 20}{9.8} = \frac{40}{9.8} = 4.08$ sec.

(e) Let R be the Horizontal range, then

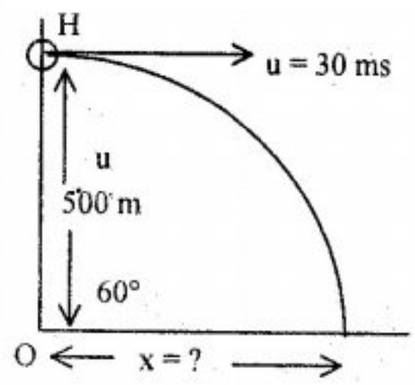
$\therefore$ Horizontal range, $H = \frac{v^2 \sin 2\alpha}{g} = \frac{2.v\cos\alpha.v\sin\alpha}{g} = \frac{2.25.20}{9.8} = 102.04$ m.

3. A helicopter flying horizontally with a speed of 30 ms^{-1} at an altitude of 500 m has to drop a food packet for a person standing on the ground. At what distance from the person should the packet be dropped ? The man stands in the vertical plane of the helicopter's motion. ($g = 10\ ms^{-2}$)

Sol^n: Let, H be the position of the helicopter at an altitude of 500 m from the ground and M be the position of the man. Since the helicopter flying horizontally, its vertical velocity component is zero. So, we have $h = \frac{1}{2} gt^2$ $\Rightarrow 500 = \frac{1}{2} \times 10 \times t^2$

$\Rightarrow \quad 10t^2 = 1000$ $\Rightarrow t^2 = 100$

$\therefore \quad t = 10$ sec



Let, x be the required distance of the dropped packet from the man.

Also, the horizontal velocity component = 30 ms^{-1}

$\therefore \quad x = ut = 30 \times 10 = 300$ m.

4. Prove that a projectile will rise three times as high when its angle of elevation is 60° as when it is 30°, but will cover the same horizontal distance.

Sol^n: Let, u be the velocity attained by the projectile.

Case I : For the height attained by the projectile

Let, H_1 and H_2 be the height attained by the projectile when angle of elevation are 60° and 30° respectively. Then

$$H_1 = \frac{u^2 \sin^2 60°}{2g} = \frac{u^2}{2g} \times \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3u^2}{8g} \quad \ldots\ldots (i)$$

$$H_2 = \frac{u^2 \sin^2 30°}{2g} = H_2 = \frac{u^2}{2g} \times \left(\frac{1}{2}\right)^2 = \frac{u^2}{8g} \quad \ldots\ldots \text{(ii)}$$

Dividing (i) by (ii), we get

$$\frac{H_1}{H_2} = \frac{3}{1} \quad \Rightarrow H_1 = 3H_2$$

$\therefore$ H_1 is the three times greater than H_2.

Case II : For horizontal range

Let, R_1 and R_2 be the horizontal range of the projectile when angle of elevation are 60° and 30° respectively. Then

$$R_1 = \frac{u^2 \sin(2 \times 60°)}{g} = \frac{u^2}{g} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}\, u^2}{2g} \quad \ldots\ldots \text{(iii)}$$

$$R_2 = \frac{u^2 \sin(2 \times 30°)}{g} = \frac{u^2}{g} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}\, u^2}{2g} \quad \ldots\ldots \text{(iv)}$$

From (iii) and (iv), we get $R_1 = R_2$

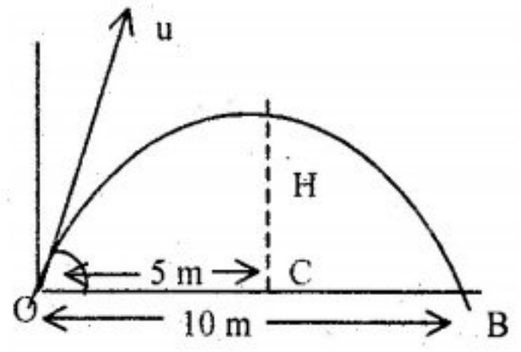
$\therefore$ Two horizontal range are same.

5. A spring gun projects a golf ball at an angle of 45° above the horizontal. The horizontal range is 10 m. ($g = 10\ ms^{-2}$)

(a) What is the maximum height to which the ball rises ?

(b) For the same initial speed, what are the two angles of departure for which the range is 5 m ?

Sol^n: Let, u be the velocity and T be the time of flight of the projectile and the horizontal range (R) = 10 m. Then



$$T = \frac{2u \sin 45°}{g}$$

and $R = u \cos\alpha \, .T$

$$\Rightarrow \quad 10 = u \cos 45° \times \frac{2u \sin 45°}{g}$$

$$\Rightarrow \quad 10 = u \times \frac{1}{\sqrt{2}} \times \frac{2u}{g} \times \frac{1}{\sqrt{2}}$$

$$\Rightarrow \quad u^2 = 10g = 10 \times 10$$

$\therefore$ $u = 10\ ms^{-1}$

(a) Let, H be the maximum height,

Then, $H = \frac{u^2 \sin^2\alpha}{2g} = \frac{10^2 \times (\sin 45°)^2}{2 \times 10} = \frac{100}{20} \times \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{5}{2} = 2.5$ m.

(b) Let, α is the angle of departure when the range 5m.

Then $R = \frac{u^2 \sin 2\alpha}{g}$

$$\Rightarrow \quad \sin 2\alpha = \frac{gR}{u^2} = \frac{10 \times 5}{10 \times 10} = \frac{1}{2}$$

$$\Rightarrow \quad \sin 2\alpha = \sin 30° \text{ and } \sin 2\alpha = \sin 150°$$

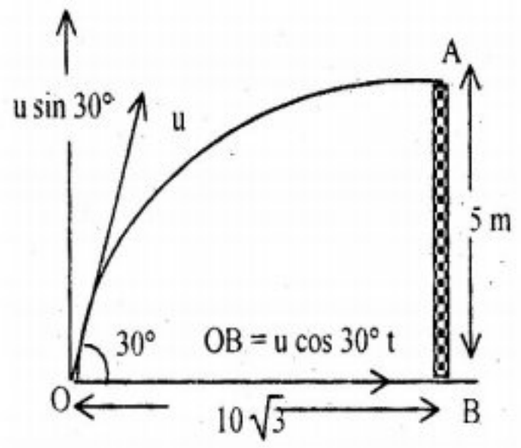
$$\Rightarrow \quad 2\alpha = 30° \text{ and } 2\alpha = 150°$$

$$\Rightarrow \quad \alpha = 15°. \text{ and } \alpha = 75°.$$

6. A ball is thrown upwards at an angle of 30° to the horizon and lands on the top edge of a building that is $10\sqrt{3}$ m away. The top edge is 5 m above the throwing point. How fast was the ball thrown ? ($g = 10\ ms^{-2}$)

Sol^n: Let, A be the top edge of the building AB. Suppose u be the velocity of the ball making the angle $\alpha = 30°$ with horizon and t be the time taken by the ball to reach the top of the building.

Also from figure, $OB = 10\sqrt{3}$, $AB = 5$



The vertical component of velocity of projection = u sin30°.

$\therefore \quad h = (u \sin 30°)\, t - \frac{1}{2} gt^2$

or, $\quad 5 = u \times \frac{1}{2} t - \frac{1}{2} \times 10 \times t^2$

$\therefore \quad ut - 10t^2 = 10$ ..... (i)

Again horizontal component of velocity of projection = u cos30°t

or, $\quad 10\sqrt{3} = ut\, \frac{\sqrt{3}}{2}$

$\therefore \quad ut = 20$ ..... (ii)

Eliminating ut from (i) and (ii) we get

$20 - 10t^2 = 10 \qquad \Rightarrow 10t^2 = 10 \qquad \Rightarrow t = 1$

Therefore, from (ii)

$u = \frac{20}{t} = \frac{20}{1} = 20\ ms^{-1}$.

7. Find the angle of projection where the range on a horizontal plane is (a) 4 (b) $4\sqrt{3}$ times the greatest height.

Sol^n: Let, α be the angle of projection, u is the velocity of the projection, R is the horizontal range and H is the greatest height attained.

We have, $H = \frac{u^2 \sin^2\alpha}{2g}$, $\qquad R = \frac{u^2 \sin 2\alpha}{g}$

(a) Here, horizontal range is 4 times the greatest height

Then, $R = 4H$

$\Rightarrow \quad \frac{u^2 \sin 2\alpha}{g} = 4 \times \frac{u^2 \sin^2\alpha}{2g}$

$\Rightarrow \quad \sin 2\alpha = 2\sin^2\alpha$

$\Rightarrow \quad 2\sin\alpha.\cos\alpha = 2\sin^2\alpha \qquad \Rightarrow \tan\alpha = 1 \qquad \therefore \alpha = 45°.$

(b) If horizontal range is $4\sqrt{3}$ times the greatest height

Then $R = 4\sqrt{3}\ H$

$\Rightarrow \quad \frac{u^2 \sin 2\alpha}{g} = 4\sqrt{3} \times \frac{u^2 \sin^2\alpha}{2g}$

$\Rightarrow \quad \sin 2\alpha = 2\sqrt{3}\ \sin^2\alpha$

$\Rightarrow \quad 2\sin\alpha.\cos\alpha = 2\sqrt{3}\ \sin^2\alpha$

$\Rightarrow \quad \tan\alpha = \frac{1}{\sqrt{3}}$

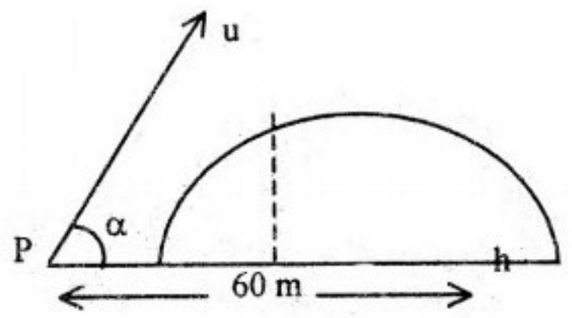
$\Rightarrow \quad \tan\alpha = \tan 30° \qquad \therefore \alpha = 30°.$

8. A projectile thrown from a point in a horizontal plane comes back to the plane in 4 secs at a distance of 60 m in front of the point of projection; find the velocity of projection. ($g = 10\ ms^{-2}$)

Sol^n: time of projection (t) = 4 sec

horizontal range (R) = 60 m

velocity of projection (u) = ?



Let, α be the angle of projection. Then we have

$$t = \frac{2u\sin\alpha}{g} \text{ and } R = \frac{u^2\sin2\alpha}{g}$$

$$\Rightarrow \quad 4 = \frac{2u\sin\alpha}{10} \text{ and } 60 = \frac{u^2 \times 2\sin\alpha\cos\alpha}{10}$$

$$\Rightarrow \quad u\sin\alpha = 20 \text{ ..... (i)} \qquad \text{and } u^2\sin\alpha\cos\alpha = 300 \text{ ..... (ii)}$$

Dividing (ii) by (i) we get

$$u\cos\alpha = 15$$

$$\text{or,} \quad \frac{20}{\sin\alpha}\cdot\cos\alpha = 15 \qquad \text{or, } \tan = \frac{4}{3} \qquad \therefore \sin\alpha = \frac{4}{5}$$

Substituting the value of $\sin\alpha$ on (i), we get

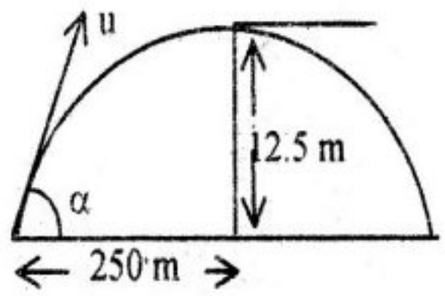
$$u\sin\alpha = 20$$

$$\text{or,} \quad u\frac{4}{5} = 20$$

$$\therefore \quad u = 25\ ms^{-1}.$$

9. Find the velocity and the direction of projection of a shot which passes in a horizontal direction just over the top of a wall which is 250 m off and 125 m high. ($g = 9.8\ ms^{-2}$)

Sol^n: Let u be the velocity, α be angle of projection of a shot. Since the shot just passes the top of the building of 125 m high.



$\therefore$ Maximum height (H) = 125 m

and horizontal range (R) = 2×250 m = 500 m

$$\therefore \quad H = \frac{u^2\sin^2\alpha}{2g} \qquad \Rightarrow 125 = \frac{u^2\sin^2\alpha}{2g} \text{ ..... (i)}$$

$$\text{and} \quad R = \frac{u^2\sin2\alpha}{g} \qquad \Rightarrow 500 = \frac{u^2\sin2\alpha}{g} \text{ ..... (ii)}$$

Dividing (i) by (ii), we get

$$\frac{1}{4} = \frac{\sin^2\alpha}{2\sin2\alpha} \qquad \Rightarrow \frac{1}{4} = \frac{\sin^2\alpha}{4\sin\alpha.\cos\alpha}$$

$$\Rightarrow \quad 1 = \frac{\sin\alpha}{\cos\alpha} \Rightarrow \tan\alpha = 1 \qquad \therefore \alpha = 45°$$

Substituting the values of $\alpha = 45°$ on (i), we get

$$125 = \frac{u^2}{2g} \times \left(\frac{1}{\sqrt{2}}\right)^2$$

$$\Rightarrow \quad u^2 = 500\ g$$

$$\Rightarrow \quad u^2 = 500 \times 9.8 = 4900$$

$$\therefore \quad u = 70\ ms^{-1}$$

10. The horizontal and vertical components of the initial velocity of a projectile are U and V respectively. If R be the range and H the greatest height attained, prove that

a) $\frac{4H}{R} = \frac{V}{U}$

Sol^n: Let, u = initial velocity of the projectile

α = angle of projection

$\therefore$ $U = u\cos\alpha$ and $V = u\sin\alpha$

We have, $H = \frac{u^2\sin^2\alpha}{2g}$ and $R = \frac{u^2\sin 2\alpha}{g}$

Now, $\frac{4H}{R} = \frac{4 \times \frac{u^2\sin^2\alpha}{2g}}{\frac{u^2\sin 2\alpha}{g}} = \frac{2u^2\sin^2\alpha}{g} \times \frac{g}{u^2\sin 2\alpha} = \frac{2\sin^2\alpha}{2\sin\alpha \,.\, \cos\alpha} = \frac{\sin\alpha}{\cos\alpha} = \frac{u\sin\alpha}{u\cos\alpha} = \frac{V}{U}$.

b) $\left(\frac{R}{U}\right)^2 = \frac{8H}{g}$

$\left(\frac{R}{U}\right)^2 = \left(\frac{u^2\sin 2\theta}{g \times u\cos\theta}\right)^2 = \left(\frac{2u\sin\theta.\cos\theta}{g \times \cos\theta}\right)^2 = \frac{4u^2\sin^2\theta}{g^2} = \frac{4}{g}\left(\frac{u^2\sin^2\theta}{2g}\right) \times 2 = \frac{8H}{g}$.

11. A projectile shot at an angle of 60° above the horizontal strikes a building $30\sqrt{3}$ m away at a point 85 m above the point of projection. ($g = 10\ ms^{-2}$)

a) Find the speed of projection.

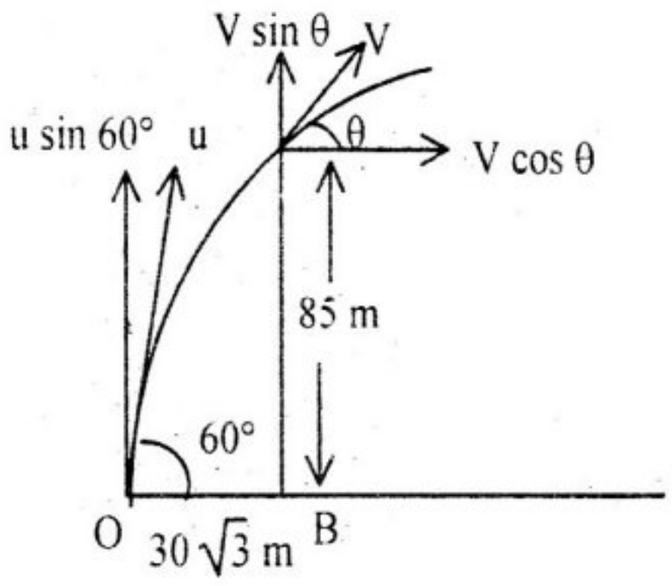
b) Find the magnitude and direction of the velocity of the projectile when it strikes the building.

Sol^n: Let u be the velocity, α be angle of projection of a shot such that $\alpha = 60°$. Here,

horizontal distance of particle = $30\sqrt{3}$ m. and

height above the point of projection = 85 m

time taken by the projectile to reach the height = t



(a) horizontal velocity component of u = ucos60°

The vertical component of velocity u = u sin60°.

$\therefore$ $h = u\sin 60° \times t - \frac{1}{2}gt^2$

$\Rightarrow$ $85 = u \times \frac{\sqrt{3}}{2}t - \frac{1}{2} \times 10 \times t^2$

$\Rightarrow$ $10t^2 = \sqrt{3}\,(ut) - 170$ ..... (i)

$\therefore$ $u\cos 60°t = 30\sqrt{3}$

$\Rightarrow$ $u \times \frac{1}{2}t = 30\sqrt{3}$

$\Rightarrow$ $ut = 60\sqrt{3}$ ..... (ii)

Eliminating ut from (ii) on (i), we get

$10t^2 = \sqrt{3} \times 60\sqrt{3} - 170$ $\Rightarrow 10t^2 = 180 - 170$ $\therefore$ $t = 1$

Substituting the value of t on (ii), we get

$u = 60\sqrt{3}\ ms^{-1}$.

(b) Let v be the striking velocity of the shot making an angle θ with horizontal.

$v_x = v\cos\theta = 60\sqrt{3}\cos60° = 60\sqrt{3} \times \frac{1}{2} = 30\sqrt{3}$

and $v_y = v\sin\theta = u\sin\alpha - gt = 60\sqrt{3}\sin60° - 10 \times 1 = 60\sqrt{3} \times \frac{\sqrt{3}}{2} - 10 = 80$

Now, $v^2 = v_x^2 + v_y^2 = (30\sqrt{3})^2 + (80)^2 = 9100$

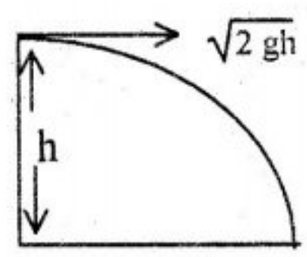
$\therefore \quad v = 10\sqrt{91}$ m/s.

Also, $\tan\theta = \frac{v_y}{v_x} = \frac{80}{30\sqrt{3}} = \frac{8}{3\sqrt{3}} \qquad \therefore \theta = \tan^{-1}\left(\frac{8}{3\sqrt{3}}\right)$.

12. A stone is thrown horizontally with velocity $\sqrt{2gh}$ from the top of a tower of height h. Find where it will strike the level ground through the foot of the tower. What will be its striking velocity ?

Sol^n: Let, t be the time taken by the stone when it strikes the ground after falling a height h.

Then, $h = 0 + \frac{1}{2}gt^2 \qquad \therefore t = \sqrt{\frac{2h}{g}}$



Here, the stone was thrown horizontally. So the angle of projection, $\alpha = 0$

$\therefore$ Horizontal distance (R) $= u\cos\alpha\, t$

$= \sqrt{2gh} \times \cos0° \times \sqrt{\frac{2h}{g}} = \sqrt{4h^2} = 2h.$

Let, v be the striking velocity of the stone at the level of foot of the tower

Then, $v^2 = u^2 + 2gh = 2gh + 2gh = 4gh = (2\sqrt{gh})^2$

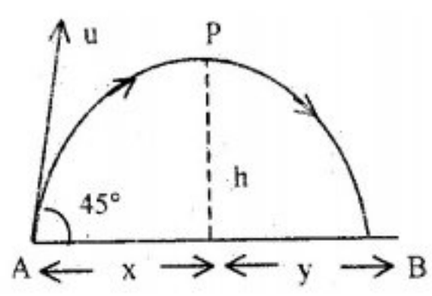
$\therefore \quad v = 2\sqrt{gh}$.

13. From a point on the ground at a distance x from the foot of a vertical wall, a ball is thrown at an angle of 45° which just clears the top of the wall and afterwards strikes the ground at a distance y on the other side. Prove that the height of the wall is $\frac{xy}{x+y}$.

Sol^n: Let initial velocity = u m/s

angle of projection (α) = 45°

Let, t be the time taken by the ball to cover the horizontal distance x.



$\therefore \quad x = u\cos45° \times t \qquad \Rightarrow t = \frac{x\sqrt{2}}{4}$

Suppose, h be the height of wall

$\therefore \quad h = (u\sin45°)\,t - \frac{1}{2}gt^2 = \frac{u}{\sqrt{2}} \times \frac{x\sqrt{2}}{u} - \frac{1}{2}g \times \frac{x^2 \times 2}{u^2} = x - \frac{x^2 g}{u^2}$ ..... (i)

Again, horizontal range $= \frac{u^2}{g}$

$\therefore \quad x + y = \frac{u^2}{g} \qquad \Rightarrow u^2 = g(x+y)$ ..... (ii)

Eliminating u^2 from (i) and (ii), we get

$h = x - \frac{x^2 g}{g(x+y)} = x - \frac{x^2}{x+y} = \frac{xy}{x+y}$.

14. If R be the horizontal range of a projectile and h its greatest height, prove that its initial velocity is $\sqrt{2g\left(h + \frac{R^2}{16h}\right)}$.

Sol^n: Let, u be the velocity of the projection and α, the angle of projection. Then,

$$\text{greatest height (h)} = \frac{u^2 \sin^2\alpha}{2g}$$

$$\text{and horizontal range (R)} = \frac{u^2 \sin 2\alpha}{g}$$

$$\text{Now, } \frac{R^2}{16h} = \frac{u^4 \,.\, (\sin 2\alpha)^2}{g^2} \times \frac{1}{16} \times \frac{2g}{u^2 \sin^2\alpha}$$

$$= \frac{u^2 \,.\, 4\sin^2\alpha \,.\, \cos^2\alpha}{g^2} \times \frac{1}{8} \times \frac{g}{u^2 \sin^2\alpha} = u^2 \times \frac{1}{2g} \cos^2\alpha \quad \ldots\ldots \text{(i)}$$

$$\therefore \quad \sqrt{2g\left(h + \frac{R^2}{16h}\right)} = \sqrt{2g\left(\frac{u^2 \sin^2\alpha}{2g} + \frac{u^2 \cos^2\alpha}{2g}\right)}$$

$$= \sqrt{u^2 (\sin^2\alpha + \cos^2\alpha)}$$

$$= \sqrt{u^2 \times 1}$$

$$= u$$

Hence, initial velocity u is $\sqrt{2g\left(h + \frac{R^2}{16h}\right)}$.

15. A cannon ball has the same range R on a horizontal plane for two different angles of projection. If H and H' are the greatest heights and t_1 and t_2 are the time of flights in two paths for which this is possible, prove that

a) $R^2 = 16HH'$ **b) $R = \frac{1}{2} gtt'$**

Sol^n: Let, α and β are two angles of projection for same horizontal range R

Then, $\frac{u^2 \sin 2\alpha}{2g} = \frac{u^2 \sin 2\theta}{2g}$

$\Rightarrow \quad \sin 2\alpha = \sin 2\theta$

$\Rightarrow \quad \sin 2\alpha = \sin 2\theta$ and $\sin(180 - 2\theta)$ $\Rightarrow \alpha = \theta$ and $90 - \theta$

Let, H be max. height t be time of flight for $\alpha = \theta$ and H' be the max. height and t' for $\alpha = 90 - \theta$

a) Now, $16\ HH' = 16 \left(\frac{u^2 \sin^2\theta}{2g}\right) \times \left(\frac{u^2 \sin^2(90 - \theta)}{2g}\right)$

$$= 4 \times \frac{u^2 \sin^2\theta \,.\, u^2 \cos^2\theta}{g^2} = 4 \frac{(2\sin\theta.\cos\theta)^2 u^2}{4g^2} = \left(\frac{u^2 \sin 2\theta}{g}\right)^2 = R^2$$

$\therefore \quad R^2 = 16\ HH'$

b) Now, $\frac{1}{2} gtt' = \frac{1}{2} g \times \left(\frac{2u\sin\theta}{g}\right) \times \left(\frac{2u\sin(90 - \theta)}{g}\right)$

$$= 2 \frac{u^2 \sin\theta \,.\, \cos\theta}{g} = \frac{u^2 \sin 2\theta}{g} = R.$$

$\therefore \quad R = \frac{1}{2} gtt'$

16. A ball is projected with a velocity of 49.0 m/s, find the two directions along which the ball must be projected so as to have a range of 122.5 m.

Sol^n: Initial velocity (u) = 49.0 m/s

Horizontal range (R) = 122.5

Angle of projection = α

We have, $R = \frac{u^2\sin^2\alpha}{g}$

$\Rightarrow \quad 122.5 = \frac{49^2}{9.8}\sin^2\alpha$

$\Rightarrow \quad \sin2\alpha = 122.5 \times \frac{9.8}{49 \times 49} = \frac{1}{2}$

$\Rightarrow \quad \sin2\alpha = \sin30^\circ$ or $\sin150^\circ$

$\Rightarrow \quad 2\alpha = 30^\circ$ or 150°

$\Rightarrow \quad \alpha = 15^\circ$ or 75°.

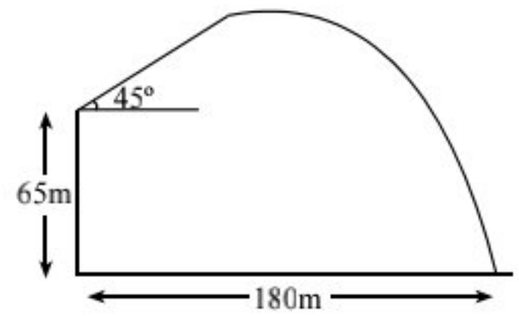
17. With what velocity must a body be projected at an angle of 45° from the top of a tower 65 m high, if it is to reach a point on the ground 180 m from the base of the tower.

Sol^n: Height of tower (h) = 65m

Horizontal range (x) = 180m

Time of flight = t sec

Angle of projection (α) = 45°



$x = u\cos\alpha.\ t$

$\Rightarrow\ 180 = u.\ \frac{1}{\sqrt{2}}\ .\ t$

$\Rightarrow \quad ut = 180\sqrt{2} \quad \ldots\ldots\ldots$ (i)

Also, taking upward direction positive

$-h = u\sin\alpha.\ t - \frac{1}{2}gt^2$

$\Rightarrow \quad -65 = 180\sqrt{2} \times \frac{1}{2} \times 9.8t^2$

$\Rightarrow \quad 4.9t^2 = 180 + 65$

$\Rightarrow \quad t^2 = \frac{245}{4.9} = 50$

$\Rightarrow \quad t = 5\sqrt{2}$ see.

$\therefore$ Putting $t = 5\sqrt{2}$, $u \times 5\sqrt{2} = 180\sqrt{2} \quad \Rightarrow u = 36$m/s .

18. A body is projected with a velocity of 16 $\sqrt{2}$ m/sec from the top of a tower 16 m high at an elevation of 45°. Find where and with what velocity will it strike the ground? (g = 10 m/s^2)

Sol^n: Initial velocity (u) = $16\sqrt{2}$ m/s

Height of travel (h) = 16m

Angle of projection (α) = 45°

Taking upward direction positive we have,

$-h = u\sin\alpha.t - \frac{1}{2}gt^2$

$\Rightarrow \quad -16 = 16\sqrt{2} \cdot \frac{1}{\sqrt{2}} \cdot t - \frac{1}{2} \times 10t^2$

$\Rightarrow \quad -16 = 16t - 5t^2$

$\Rightarrow \quad 5t^2 - 16t - 16 = 0$

$\Rightarrow \quad 5t^2 - 20t + 4t - 16 = 0$

$\Rightarrow \quad 5t(t-4) + 4(t-4) = 0$

$\Rightarrow \quad (t-4)(5t+4) = 0$

$\Rightarrow \quad t = 4, -\frac{4}{5}$

Taking positive, t = 4 sec.

Horizontal distance (x) $= u\cos\alpha . t = 16\sqrt{2} \times \frac{1}{\sqrt{2}} \times 4 = 6400$ m.

Let v be the velocity of striking the ground and θ the angle made with horizontal.

$$v_x = v\cos\theta = u\cos\alpha = 16\sqrt{2} \times \frac{1}{\sqrt{2}} = 16 \text{m/s}$$

and $v_y = v\sin\theta = -u\sin\alpha + gt = -16\sqrt{2} \times \frac{1}{\sqrt{2}} + 10 \times 4 = -16 + 40 = 24$ m/s

Now, $v = \sqrt{v_x^2 + v_y^2} = \sqrt{16^2 + 24^2} = \sqrt{256 + 576} = \sqrt{832} = 8\sqrt{13}$ m/s

Now, $\tan\theta = \frac{v_y}{y_x} = \frac{24}{16} = \frac{3}{2} \qquad \Rightarrow \theta = \tan^{-1}\left[\frac{3}{2}\right]$.

Hint and Solution of MCQ's

1. The way of a projectile in the air may in any direction.
2. The path of projectile is parabolic and it is called trajectory.
3. The angle made by the direction of the projected body with the horizontal line is angle of projection.
4. At the point, where a projectile attains its maximum height has horizontal velocity is zero.
5. Path of projectile, $y = bx - ax^2$

   Comparing with $y = Ax^2 + Bx + c$ we get

   $A = -a, B = b, C = 0$

   The greatest height attains at $x = -\frac{B}{2A} = \frac{b}{2a}$.

   When $x = \frac{b}{2a}$, $y = b.\frac{b}{2a} - a.\frac{b^2}{4a^2} = \frac{b^2}{2a} - \frac{b^2}{4a} = \frac{2b^2 - b^2}{4a} = \frac{b^2}{4a}$
6. Time of flight $= \frac{2u\sin\alpha}{g}$

   $\therefore$ time to reach greatest height $= \frac{1}{2} \cdot \frac{2u\sin\alpha}{g} = \frac{u\sin\alpha}{g}$
7. $R_{max} = \frac{u^2 \sin 2\alpha}{g}$

   Which will be maximum when $\sin2\alpha$ is max.

   $\therefore$ max. $\sin2\alpha = 1 \qquad \Rightarrow 2\alpha = 90^\circ \qquad \Rightarrow \alpha = 45^\circ$
8. Let u be the velocity of projection, R the given range and α the angle of projection. Then

   $$R = \frac{u^2 \sin 2\alpha}{g} = \frac{u^2 \sin(\pi - 2\alpha)}{g} = \frac{u^2 \sin 2\left(\frac{\pi}{2} - \alpha\right)}{g} = \frac{u^2 \sin 2\beta}{g}$$

   $\therefore$ Two direction of projection for same horizontal range is α and $\beta = \frac{\pi}{2} - \alpha$.

9. If one direction of projectile, $\alpha = \frac{\pi}{6}$ then next direction $= \frac{\pi}{2} - \alpha = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$.

10. $R_{max} = \frac{u^2 \sin 2\alpha}{g}$ will be greatest for $\sin 2\alpha = 1$

$\therefore \quad R_{max} = \frac{u^2}{g}$.

11. $\alpha = 30^o$, $u = 40$m/s

time taken to reach maximum height $= \frac{u \sin\alpha}{g} = \frac{40.\ \sin 30^o}{10} = 2$ s

12. $h = -u_y t + \frac{1}{2} g t^2$ (taking upward direction positive)

$\Rightarrow \quad 60 = -u \sin\alpha \times 4 + \frac{1}{2} \times 10 \times 4^2$

$\Rightarrow \quad 60 = -40 \sin\alpha + 80$

$\Rightarrow \quad 40 \sin\alpha = 20 \quad \Rightarrow \sin\alpha = \frac{1}{2} \quad \Rightarrow \alpha = 30^o$

13. $R_{max} = \frac{u^2}{g} = \frac{20^2}{10} = 40$m.

14. $R_{max} = \frac{u^2}{g} = 100$m for $\alpha = 45^o$

$\therefore \quad H_{max} = \frac{u^2 \sin^2\alpha}{2g} = \frac{u^2}{g} . \frac{\sin^2 45^o}{2} = 100 . \frac{1}{4} = 25$m

15. $R = \frac{u^2 \sin 2\alpha}{g}$

$\Rightarrow \quad 20 = \frac{20^2 . \sin 2\alpha}{10} \quad \Rightarrow \sin 2\alpha = \frac{1}{2}$

$\Rightarrow \quad 2\alpha = 30^o \quad \Rightarrow \alpha = 15^o$

$\therefore$ two direction of projection are α and $90^o - \alpha = 15^o$ and 75°

16. $R = 4H$

$\Rightarrow \quad \frac{u^2 \sin 2\alpha}{g} = 4 . \frac{u^2 \sin^2\alpha}{2g}$

$\Rightarrow \quad 2\sin\alpha\cos\alpha = 2 \sin^2\alpha \quad \Rightarrow \tan\alpha = 1 \quad \Rightarrow \alpha = 45^o$

17. $\alpha = 45^o$, $t = 4\sqrt{2}$ s , $u = ?$

Since the projectile passes a wall of height h horizontally, h is greatest height

And time taken to reach greatest height (t) $= \frac{u \sin\alpha}{g}$

$\Rightarrow \quad 4\sqrt{2} = \frac{u . \sin 45^o}{10}$

$\Rightarrow \quad 4\sqrt{2} = \frac{u}{10\sqrt{2}}$

$\Rightarrow \quad u = 80$m/s.